

Unit : 3 Linear transformation

Qm:

1) Define Linear Transformation.

Sol:

Let V and W be vector over the field F .

A function $T : V \rightarrow W$ is called as linear transformation if $\forall x, y \in V$ and $c \in F$

$$(i) T(x+y) = T(x) + T(y)$$

$$(ii) T(cx) = cT(x)$$

$$(or) [(x+y) + (cx)] T = [T(x) + T(y) + cT(x)]$$

$$T(\alpha x + y) = \alpha T(x) + T(y) \quad \forall x, y \in V \text{ and } \alpha \in \mathbb{R}.$$

2) If $T : V \rightarrow W$ is linear transformation then show that $T(x-y) = T(x) - T(y)$

Sol: Let $x, y \in V$

$$(i) \Rightarrow [T(x), T(y)] \in W$$

$$\text{consider } T(x-y) = T(x+(-y))$$

$$\begin{aligned} &= T(x) + T(-y) \\ &= T(x) + T[(-1)y] \\ &= T(x) + (-1)T(y) \\ &= T(x) - T(y) \end{aligned}$$

$$\therefore LHS = RHS.$$

Qm:

1) ii) Define reflection about x -axis and projection on x -axis and show that they are linear transformations.

Q01: Reflection about x axis

Define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$T(a_1, a_2) = (a_1, -a_2)$ is called reflection about the x-axis.

To prove: Reflection on x axis is linear transformation

Proof: Let $x, y \in \mathbb{R}^2, c \in \mathbb{R}$

$$\therefore x = (a_1, a_2), y = (b_1, b_2)$$

Now,

$$\text{LHS: } T[cx+y] = T[c(a_1, a_2) + (b_1, b_2)]$$

$$= T[(ca_1, ca_2) + (b_1, b_2)]$$

$$= T[ca_1 + b_1, ca_2 + b_2]$$

$$\stackrel{\text{from definition}}{=} [ca_1 + b_1, -(ca_2 + b_2)]$$

$$\therefore \boxed{T[cx+y] = cT(x) - T(y)}$$

$$= [ca_1 + b_1, -(ca_2 + b_2)] \rightarrow \textcircled{1}$$

RHS:

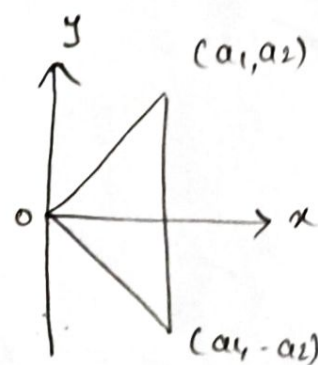
$$cT(x) + T(y)$$

$$= cT(a_1, a_2) + T(b_1, b_2)$$

$$= (c(a_1, -a_2)) + (b_1, b_2)$$

$$= [ca_1, -ca_2] + (b_1, b_2)$$

$$= [ca_1 + b_1, -ca_2 + b_2] \rightarrow \textcircled{2}$$



Reflection about x-axis

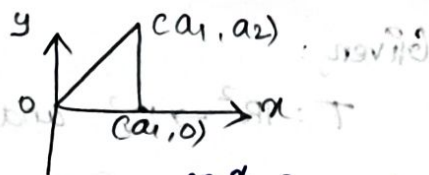
From $\textcircled{1}$ and $\textcircled{2}$, we get LHS = RHS

$$\text{(ie)} \quad T(cx+y) = cT(x) + T(y)$$

$\therefore T$ is linear transformation.

Projection on x axis:

Define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by



$T(a_1, a_2) = (a_1, 0)$ is called projection on x axis

To prove projection on x axis is linear transformation

Proof let $x, y \in \mathbb{R}^2, c \in \mathbb{R}$

$$\therefore x = (a_1, a_2) \quad y = (b_1, b_2)$$

Now:

$$\begin{aligned} \text{LHS } T(cx+y) &= T[c(a_1, a_2) + (b_1, b_2)] \\ &= T[(ca_1, ca_2) + (b_1, b_2)] \\ &= T[ca_1 + b_1, ca_2 + b_2] \\ &= [ca_1 + b_1, 0] \rightarrow \textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{RHS } [cT(x) + T(y)] &= [cT(a_1, a_2) + T(b_1, b_2)] \\ &= [c(a_1, 0) + (b_1, 0)] \\ &= [(ca_1, 0) + (b_1, 0)] \\ &= (ca_1 + b_1, 0) \rightarrow \textcircled{2} \end{aligned}$$

From $\textcircled{1}$ & $\textcircled{2}$ we get

$$T(cx+y) = cT(x) + T(y)$$

$\therefore$ The projection on x axis is linear transformation

(iv) Check if $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as given below, are linear transformations or not.

(a) $T(a_1, a_2) = (a_1, a_2^2)$

(b) $T(a_1, a_2) = (a_1 + 1, a_2)$

$$\begin{aligned} [(ca_1, ca_2) + (b_1, b_2)] T &= (c+1, c+1) T \quad \text{LHS} \\ [(ca_1, ca_2) + (b_1, b_2)] T &= \end{aligned}$$

Given:

$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and defined as $T(x) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} x$

a) $T(a_1, a_2) = (a_1, a_2)$ is $(0, 1) = (0, 1)$ and $(1, 0) = (1, 0)$

To check: $T(cx + y) = cT(x) + T(y)$

$\forall x, y \in \mathbb{R}^2$ and $c \in \mathbb{R}$

Let $x = (a_1, a_2)$

$y = (b_1, b_2)$

LHS:

$$= T(cx + y)$$

$$= T[c(a_1, a_2) + (b_1, b_2)]$$

$$= T[(ca_1, ca_2) + (b_1, b_2)]$$

$$= T[(ca_1 + b_1, ca_2 + b_2)]$$

$$= [ca_1 + b_1, ca_2 + b_2]$$

$$= [ca_1 + b_1, c^2 a_2^2 + 2b_2 ca_2 + b_2^2] \rightarrow \textcircled{1}$$

RHS:

$$= cT(x) + T(y)$$

$$= cT(a_1, a_2) + T(b_1, b_2)$$

$$= c(a_1, a_2) + (b_1, b_2)$$

$$= (ca_1, ca_2) + (b_1, b_2)$$

$$= (ca_1, ca_2) + (b_1, b_2) \rightarrow \textcircled{2}$$

$$= (ca_1 + b_1, ca_2 + b_2) \rightarrow \textcircled{2}$$

From $\textcircled{1}$ and $\textcircled{2}$ we got

LHS $\neq$ RHS

$\therefore$ (a) is not a linear transformation

b) Given:

$$T(a_1, a_2) = (a_1 + 1, a_2)$$

LHS $T(cx + y) = T[c(a_1, a_2) + (b_1, b_2)]$
 $= T[(ca_1, ca_2) + (b_1, b_2)]$

$$= T[ca_1 + b_1, ca_2 + b_2]$$

$$= (ca_1 + b_1 + 1, ca_2 + b_2) \rightarrow \textcircled{3}$$

RHS:

$$cT(x) + T(y) = cT(a_1, a_2) + T(b_1, b_2)$$

$$= c(a_1, a_2) + (b_1, b_2)$$

$$= (ca_1 + 1, a_2) + (b_1 + 1, b_2)$$

$$= (ca_1 + c + b_1 + 1, ca_2 + b_2)$$

$$= [ca_1 + c + b_1 + 1, ca_2 + b_2] \rightarrow \textcircled{4}$$

From $\textcircled{3}$ and $\textcircled{4}$

we get

$$\text{LHS} \neq \text{RHS}$$

$\therefore$ (b) is not a linear transformation //

(i) Let V and W be two vector spaces over a field F . Define zero transformation and identity transformation and prove that they are linear.

Sol:

Definition: Zero transformation

Let V and W be two vector spaces over F . Then the function $T: V \rightarrow W$ defined by $T(v) = 0 \forall v \in V$ is called zero transformation from V to W .

To prove zero transformation is linear

Proof: $\forall x, y \in V, c \in F$, we have $T(x) = 0, T(y) = 0$

By definition

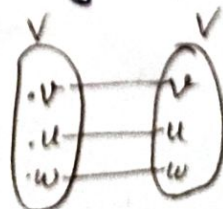
To prove: $T(cx + y) = cT(x) + T(y)$

$$\text{LHS: } T(cx + y) = 0$$

$$= 0 + 0$$

$$= c \cdot 0 + 0$$

$$= cT(x) + T(y)$$



$$\text{LHS} = \text{RHS}$$

$\therefore T$ is linear transformation

Definition: Identity Transformation.

Let V be a vector space over F , then the function

$I: V \rightarrow V$ defined by $I(v) = v, \forall v \in V$ is

called Identity transformation.

To prove Identity transformation is linear

Proof: $\forall x, y \in V, c \in F \Rightarrow I(x), I(y) = y$

To prove: $I(cx+y) = cI(x) + I(y)$

$$\begin{aligned} \text{LHS: } I(cx+y) &= cx+y \\ &= cI(x) + I(y) \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS}$$

$\therefore I$ is linear transformation

(iii) If $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is defined by $T(a_1, a_2) =$

$(2a_1 + a_2, a_1)$, then prove that T is not a linear transformation.

sol:

Let $x, y \in \mathbb{R}^2, c \in \mathbb{R}$

$$\therefore x = (a_1, a_2) \quad y = (b_1, b_2)$$

To prove: $T(cx+y) = cT(x) + T(y)$

$$\text{LHS: } T(cx+y) = T[c(a_1, a_2) + (b_1, b_2)]$$

$$= T[(ca_1, ca_2) + (b_1, b_2)]$$

$$= T[ca_1 + b_1, ca_2 + b_2]$$

$$= [2(ca_1 + b_1) + ca_2 + b_2, ca_1 + b_1] \neq \text{RHS}$$

RHS:

$$T(ax) + T(y) = T(ax + y) = T(a_1, a_2) + T(b_1, b_2)$$

$$= [2a_1 + a_2, a_1] + [2b_1 + b_2, b_1]$$

$$= [2ca_1 + a_2c, ca_1] + [2cb_1 + b_2c, cb_1]$$

$$= [2ca_1 + a_2c + 2cb_1 + b_2c, ca_1 + cb_1]$$

$$= [2c(ca_1 + b_1) + ca_2 + b_2c, ca_1 + cb_1] \rightarrow \textcircled{2}$$

From $\textcircled{1}$ and $\textcircled{2}$: $[T(ax) + T(y)] = [T(ax + y)]$

LHS = RHS

$\therefore T$ is linear Transformation.

(v) If $T: M_{m \times n}(\mathbb{R}) \rightarrow M_{n \times m}(\mathbb{R})$ is defined as $T(A) = A^t$, where A^t is the transpose of A , then prove that T is a linear transformation.

Proof: Let $A, B \in M_{m \times n}(\mathbb{R})$ and $\alpha \in \mathbb{R}$, such that $(\Rightarrow) T(A) = A^t, T(B) = B^t$,

To prove:

$$\begin{aligned} T(\alpha A + B) &= \alpha T(A) + T(B) \\ \text{LHS: } T(\alpha A + B) &= (\alpha A + B)^t \\ &= (\alpha A)^t + (B)^t \\ &= \alpha A^t + B^t \\ &= \alpha T(A) + T(B) \end{aligned} \quad \left\{ \begin{array}{l} (x^n)' = nx^{n-1} \\ (x^3)' = 3x^2 \end{array} \right.$$

LHS = RHS

$\therefore T$ is linear transformation.

(vi) If $T: P_n(\mathbb{R}) \rightarrow P_{n-1}(\mathbb{R})$ is defined as $T(f(x)) = f'(x)$, where $f'(x)$ denotes the derivative of $f(x)$, then show that T is linear.

Sol:

Proof: Let $f(x) + g(x) \in P_n(\mathbb{R})$ and $c \in \mathbb{R}, \exists$

$$T[f(x)] = f'(x) \text{ and } T[g(x)] = g'(x)$$

To prove:

$$T[cf(x) + g(x)] = cT[f(x)] + T[g(x)]$$

LHS:

$$T[cf(x) + g(x)] = [cf(x) + g(x)]'$$

$$= [cf(x)]' + [g(x)]'$$

$$= cf'(x) + g'(x)$$

$$= c[T[f(x)] + T[g(x)]]$$

$$\text{LHS} = \text{RHS}$$

$\therefore T$ is linear transformation.

(vii) Let $V = C(\mathbb{R})$, the vector space of continuous real-valued functions on $\mathbb{R}$. Let $a, b \in \mathbb{R}, a < b$. Define $T: V \rightarrow \mathbb{R}$ by $T(f) = \int_a^b f(t) dt$ for all $f \in V$, then prove that T is a linear transformation.

Qm:

4) Define range and null space of a linear transformation.

Sol: Let V, W be vector spaces over the field F and $T: V \rightarrow W$ be a linear transformation.

(i) Range: Range of T is defined as the set of all images of vectors in V .

$$R(T) = \{T(x) \mid x \in V\}$$

(ii) Null space: Null space of T is the set of all vectors in V whose image is zero.

$$N(T) = \{x \in V \mid T(x) = 0\}$$

5) Define rank and nullity of linear transformation

Sol: Let V and W be vector spaces over the field F , $T: V \rightarrow W$ be a linear transformation. If $R(T)$ and $N(T)$ are finite dimensional, then

* Rank of T is defined as $\text{rank}(T) = \dim(R(T))$

* Nullity of T is defined as $\text{nullity}(T) = \dim(N(T))$

6) State dimension theorem

Sol: Let V & W be vector spaces over the field F and $T: V \rightarrow W$ be a linear transformation. If V is finite dimensional, then $\text{rank}(T) + \text{nullity}(T) = \dim(V)$

7) Define one to one linear transformation.

Sol: Let V & W be vector spaces over the field F and $T: V \rightarrow W$ be a linear transformation. If for $x, y \in V$, $T(x) = T(y) \Rightarrow x = y$ then T is said to be one-to-one linear transformation.

8m:

2) (i) If $T: V \rightarrow W$ is a linear transformation, where V and W are vector spaces, then prove that T is one-one iff and only iff $N(T) = \{0\}$.

Sol: Given that T is 1-1

i.e. (i) $T: V \rightarrow W$ is linear

$$T(ax + y) = aT(x) + T(y) \quad \forall x, y \in V, a \in F$$

(ii) T is 1-1.

$$\text{i.e. } T(x) = T(y)$$

$$\Rightarrow x = y \quad (\text{or}) \quad x \neq y \Rightarrow T(x) \neq T(y)$$

We need to prove

$$N(T) = \{0\}$$

Let $x \in N(T)$, then by definition of $N(T)$, $T(x) = 0$

$$\Rightarrow T(x) = T(0)$$

$$\Rightarrow x = 0$$

$$\Rightarrow N(T) = \{0\}$$

Given that T is linear with $N(T) = \{0\}$

To prove: T is 1-1

$$\text{i.e. } T(x) = T(y)$$

$$\Rightarrow x = y$$

$$\text{Let } T(x) = T(y)$$

$$T(x) - T(y) = 0$$

$$T(x - y) = 0 \quad (\because T \text{ is linear})$$

$$\Rightarrow x - y \in N(T) = \{0\}$$

By def of $N(T)$

$$\Rightarrow x - y = 0$$

$$\Rightarrow x = y$$

$$\therefore T \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} T = (1 \ 1 \ 1) T$$

ii) Let V and W be vector spaces and $T: V \rightarrow W$ be a linear transformation, then prove that $N(T)$ and $R(T)$ are subspaces of V and W respectively.

Sol:

Rank Nullity Dimension Theorem (RND)

Let V and W be vector space and $T: V \rightarrow W$ be linear. If T is finite dimensional, then $\text{rank}(T) + \text{nullity}(T) = \dim(V)$.

1bm:

3(i) prove that $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$ is a linear transformation. Also find basis for $N(T)$ and $R(T)$. Compute nullity and rank of T and verify dimension theorem. Also check if T is one-one or onto.

Sol: Given:

$T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ such that $T(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$

Verification of T is linear transformation

i.e., to verify $T(ax + y) = aT(x) + T(y); \forall x, y \in V = \mathbb{R}^3$

Let $x = (a_1, a_2, a_3), y = (b_1, b_2, b_3) \in \mathbb{R}^3$
and $a \in \mathbb{R} = F$

LHS:

$$\begin{aligned}T(ax+by) &= T[a(a_1, a_2, a_3) + (b_1, b_2, b_3)] \\&= T[(aa_1, aa_2, aa_3) + (b_1, b_2, b_3)] \\&= T[(aa_1+b_1), (aa_2+b_2), (aa_3+b_3)] \\&= [aa_1+b_1 - aa_2-b_2, 2(aa_3+b_3)] \\&= [a(a_1-a_2) + (b_1-b_2), 2(aa_3+b_3)] \rightarrow \textcircled{1}\end{aligned}$$

RHS:

$$\begin{aligned}aT(x) + T(y) &= aT(a_1, a_2, a_3) + T(b_1, b_2, b_3) \\&= a[a_1 - a_2, 2a_3] + [b_1 - b_2, 2b_3] \\&= [aa_1 - aa_2, 2aa_3] + [b_1 - b_2, 2b_3] \\&= [aa_1 - aa_2 + b_1 - b_2, 2aa_3 + 2b_3] \\&= [a(a_1 - a_2) + (b_1 - b_2), 2(aa_3 + b_3)] \rightarrow \textcircled{2}\end{aligned}$$

∴ from $\textcircled{1}$ and $\textcircled{2}$ we get LHS = RHS

∴ T is linear Transformation //

To find: Basis of $N(T)$

$$N(T) = \{(a_1, a_2, a_3) \in \mathbb{R}^3 \mid T(a_1, a_2, a_3) = (0, 0)\}$$

0 element $\mathbb{R}^2$ is $(0, 0)$

$$= \{ (a_1, a_2, a_3) \in \mathbb{R}^3 / (a_1 - a_2, 2a_3) = (0, 0) \}$$

$$= \{ (a_1, a_2, a_3) \in \mathbb{R}^3 / (a_1 = a_2, a_3 = 0) \}$$

$$= \{ (a_1, a_1, 0) \in \mathbb{R}^3 \}$$

$$N(T) = \{ a_1 (1, 1, 0) \in \mathbb{R}^3 / a_1 \in F \}$$

$$\text{span}[N(T)] = \text{span} \{ (1, 1, 0) \in \mathbb{R}^3 \} \text{ and}$$

$\{(1, 1, 0)\}$ is singleton set is always linearly independent.

The basis of $N(T)$ is $\{(1, 1, 0)\}$

$$\Rightarrow \dim[N(T)] = \text{Nullity of } T = 1$$

To find : Basis of $R(T)$

$$R(T) = \{ \text{Image of basis of } v \in \mathbb{R}^3 \text{ by } T \}$$

$$= [\text{Image of basis } \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\} \text{ by } T]$$

$$= \{ T(1, 0, 0), T(0, 1, 0), T(0, 0, 1) \}$$

$$= \{ (1-0, 2(0)), (0-1, 2(0)), (0-0, 2(1)) \}$$

$$= \{ (1, 0), (-1, 0), (0, 2) \}$$

$$= \{ (1, 0), (-1, 0), 2(0, 1) \}$$

$$\text{span } R(T) = \text{span} \{ (1, 0), -1(1, 0), 2(0, 1) \}$$

$$= \{ (1, 0), (0, 1) \} \text{ and } (1, 0), (0, 1) \text{ are}$$

The basis of $R(T)$ is $\{(1, 0), (0, 1)\}$ linearly independent vector

$$\Rightarrow \dim[R(T)] = \text{Rank}(T) = 2$$

Verification of RND

$$\text{Rank}(T) + \text{Nullity}(T) = \dim V$$

$$2+1 = 3$$

$$3 = 3$$

$\therefore$ RND is verified

$$\therefore \dim(N(T)) = 1 \Rightarrow N(T) \neq \{0\} \quad \boxed{T: V \rightarrow W}$$

$\Rightarrow T$ is not 1-1

$$\text{But } \dim[R(T)] = \dim W = \dim \mathbb{R}^2$$

$$\Rightarrow \dim[R(T)] = \dim \mathbb{R}^2 = 2$$

$\therefore T$ is onto

(ii) prove that $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by:

$$T(a_1, a_2) = (a_1 + a_2, 0, 2a_1 - a_2) \text{ is a}$$

linear transformation. Also find basis for $N(T)$ and $R(T)$. Compute nullity and rank of T and verify dimension theorem.

Also check if T is one-one or onto.

Sol: To prove: T is linear

$$\text{Let } u = (a_1, a_2) \in \mathbb{R}^2 \quad v = (b_1, b_2) \in \mathbb{R}^2$$

$$T(\alpha u + v) = \alpha T(u) + T(v)$$

$$\underline{\text{LHS}} \quad T(\alpha u + v) = T[\alpha(a_1, a_2) + (b_1, b_2)]$$

$$= T[(\alpha a_1, \alpha a_2) + (b_1, b_2)]$$

$$= T[(\alpha a_1 + b_1, \alpha a_2 + b_2)]$$

$$\begin{aligned}
 &= [\alpha a_1 + b_1 + \alpha a_2 + b_2, 0, 2(\alpha a_1 + b_1) - \alpha a_2 - b_2] \\
 &= [\alpha(a_1 + a_2) + (b_1 + b_2), 0, 2(\alpha a_1 + b_1) - \alpha a_2 - b_2] \\
 &\quad \rightarrow \textcircled{1}
 \end{aligned}$$

RHS:

$$\alpha T(u) + T(v) = \alpha T(a_1, a_2) + T(b_1, b_2)$$

$$\begin{aligned}
 &= \alpha [a_1 + a_2, 0, 2a_1 - a_2] + [b_1 + b_2, 0, 2b_1 - b_2] \\
 &= [\alpha(a_1 + a_2), 0, 2\alpha a_1 - \alpha a_2] + [b_1 + b_2, 0, 2b_1 - b_2] \\
 &= [\alpha(a_1 + a_2) + (b_1 + b_2), 0, 2(\alpha a_1 + b_1) - \alpha a_2 - b_2] \\
 &\quad \rightarrow \textcircled{2}
 \end{aligned}$$

From $\textcircled{1}$ and $\textcircled{2}$ we get LHS = RHS

$\therefore T$ is linear Transformation //

To find: Basis of $N(T)$

$$\begin{aligned}
 N(T) &= \{(a_1, a_2) \in \mathbb{R}^2 \mid T(a_1, a_2) = (0, 0, 0)\} \\
 &= \{(a_1, a_2) \in \mathbb{R}^2 \mid [a_1 + a_2, 0, 2a_1 - a_2] = (0, 0, 0)\} \\
 &= \{(a_1, a_2) \in \mathbb{R}^2 \mid a_1 + a_2 = 0, 0 = 0, 2a_1 - a_2 = 0\} \\
 &= \{(a_1, a_2) \in \mathbb{R}^2 \mid a_1 = -a_2, 0, 2a_1 = a_2\} \\
 &= \{(a_1, a_2) \in \mathbb{R}^2 \mid a_1 = 0, a_2 = 0\} \\
 &= \{(0, 0) \in \mathbb{R}^2\}
 \end{aligned}$$

$\text{Span}[N(T)] = \{0\}$ $(0, 0)$ is linearly Independent.

$\therefore$ Nullity of $T = \dim N(T) = 0$

To find: Basis of $R(T)$

$$R(T) = \{\text{Image of basis of } \mathbb{R}^2\}$$

$$= \{\text{Image of } (1, 0), (0, 1)\}$$

$$\text{Span } R(T) = \text{Span}\{T(1, 0), T(0, 1)\}$$

$$= \text{Span}\{(1, 0, 2), (1, 0, -1)\}$$

To prove: $\{(1, 0, 2), (1, 0, -1)\}$ is linearly independent

(or) not

let

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 0 & -3 \end{bmatrix} R_2 \rightarrow R_2 - R_1$$

It is in Echelon form but there is no zero rows.

Hence $\{(1, 0, 2), (1, 0, -1)\}$ is L.I. set.

Hence the basis of $R(T) = \{(1, 0, 2), (1, 0, -1)\}$

$$\therefore \text{Rank}(T) = \dim(R(T)) = 2$$

To verify:

RND Theorem

$$\dim N(T) + \dim[R(T)] = \dim V$$

$$0 + 2 = 2$$

$$2 = 2$$

Hence verified //

$$\text{Since } N(T) = \{0\}$$

$$\Rightarrow T \text{ is } 1-1 \text{ and } \dim[R(T)] = 2 \neq \dim W \\ = \dim \mathbb{R}^3 = 3$$

Hence T is not onto //

(ii) Prove that $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(a, b) = (a+b, a-b, b)$ is a linear transformation.

Also find basis for $N(T)$ and $R(T)$.

Compute nullity and rank of T and

Verify dimension theorem. Also check if T is one-one or onto.

Sol:

To prove that: T is linear.

let $u = (a_1, b_1)$ & $v = (a_2, b_2)$

LHS:

$$\begin{aligned} T(\alpha u + v) &= T[\alpha(a_1, b_1) + (a_2, b_2)] \\ &= T[(\alpha a_1, \alpha b_1) + (a_2, b_2)] \\ &= T[\alpha a_1 + a_2, \alpha b_1 + b_2] \\ &= (\alpha a_1 + a_2 + \alpha b_1 + b_2, \alpha a_1 + a_2 - \alpha b_1 - b_2, \\ &\quad \alpha b_1 + b_2) \rightarrow \textcircled{1} \end{aligned}$$

RHS:

$$\alpha T(u) + T(v) = \alpha T(a_1, b_1) + T(a_2, b_2)$$

$$= \alpha [a_1 + b_1, a_1 - b_1, b_1] +$$

$$[a_2 + b_2, a_2 - b_2, b_2]$$

$$= [\alpha a_1 + \alpha b_1, \alpha a_1 - \alpha b_1, \alpha b_1] +$$

$$[a_2 + b_2, a_2 - b_2, b_2]$$

$$= [\alpha a_1 + \alpha b_1 + a_2 + b_2, \alpha a_1 - \alpha b_1 + a_2 - b_2, \\ \alpha b_1 + b_2] \rightarrow \textcircled{2}$$

From $\textcircled{1}$ and $\textcircled{2}$ we get LHS = RHS

$\therefore T$ is linear.

Base of $N(T)$

$$N(T) = \{(a_1, b_1) \in \mathbb{R}^2 \mid T(a_1, b_1) = (0, 0, 0)\}$$

$$= \{(a_1, b_1) \in \mathbb{R}^2 \mid (a_1 + b_1, a_1 - b_1, b_1) = (0, 0, 0)\}$$

$$\text{Span of } N(T) = \text{Span} \{(a_1, b_1) \in \mathbb{R}^2 \mid a_1 + b_1 = 0, a_1 - b_1 = 0, b_1 = 0\}$$

$$= \text{Span} \{(a_1, b_1) \in \mathbb{R}^2 \mid a_1 = 0, b_1 = 0\}$$

$$= \text{Span} \{(0, 0) \in \mathbb{R}^2\}$$

$$\therefore \{0, 0\} \text{ is L.T.}$$

Single element is always L.I

$$\therefore \text{Basis of } N(T) = \{(0,0)\}$$

$$\text{Nullity of } T = \dim N(T) = 0$$

Basis of $R(T)$

$$R(T) = \{\text{Image of basis of } V = \mathbb{R}^2\}$$

$$R(T) = \{\text{Image} [(1,0), (0,1)]\}$$

$$\text{Span } R(T) = \text{Span}\{T(1,0), T(0,1)\}$$

$$= \text{Span}\{(1,1,0), (1,-1,1)\}$$

To check:

$$(1,1,0), (1,-1,1) \in \mathbb{R}^3 \text{ L.I or not}$$

$$\text{Let } A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \quad R_2 \rightarrow R_2 - R_1$$

It is an Echelon form but there is no zero rows.

Hence it is L.I

$$\therefore \text{Basis of } R(T) = \{(1,1,0), (1,-1,1)\}$$

$$\text{Hence Rank}(T) = \dim(R(T)) = 2$$

To Verify RND Theorem

$$\dim V = \dim \mathbb{R}^2 = 2$$

$$\text{Nullity}(T) + \text{Rank}(T) = \dim V$$

$$0 + 2 = 2$$

$$2 = 2$$

Hence verified.
 $\therefore T$ is 1-1, since $N(T) = \{0\}$ and T is not onto, since

$$\dim(R(T)) \neq \dim W = \dim \mathbb{R}^3$$

(i.e.) $2 \neq 3$.

(v) prove that $T: M_{2 \times 3}(\mathbb{R}) \rightarrow M_{2 \times 3}(\mathbb{R})$ defined by

$$T \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = \begin{bmatrix} 2a_{11} - a_{12} & a_{13} + 2a_{12} \\ 0 & 0 & 0 \end{bmatrix}$$

is linear transformation. Also find basis for $N(T)$ and $R(T)$: compute nullity and rank of T and verify dimension theorem. Also check if T is one-one or onto.

Sol: To prove: T is linear

Let $x, y \in M_{2 \times 3}(\mathbb{R})$ and $c \in \mathbb{R}$ and

$$x = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \quad y = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$$

$$\text{Then } T(x) = \begin{bmatrix} 2a_{11} - a_{12} & a_{13} + 2a_{12} \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and}$$

$$T(y) = \begin{bmatrix} 2b_{11} - b_{12} & b_{13} + 2b_{12} \\ 0 & 0 & 0 \end{bmatrix}$$

Ans: $T(cx + y) \Rightarrow$ Let $cx + y = c \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$

$$= \begin{bmatrix} ca_{11} & ca_{12} & ca_{13} \\ ca_{21} & ca_{22} & ca_{23} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$$

$$T(\alpha x + y) = T \begin{bmatrix} \alpha a_{11} + b_{11} & \alpha a_{12} + b_{12} & \alpha a_{13} + b_{13} \\ \alpha a_{21} + b_{21} & \alpha a_{22} + b_{22} & \alpha a_{23} + b_{23} \end{bmatrix}$$

$$= \begin{bmatrix} 2(\alpha a_{11} + b_{11}) - \alpha a_{12} - b_{12} & \alpha a_{13} + b_{13} + 2(\alpha a_{12} + b_{12}) \\ 0 & \dots \end{bmatrix} \rightarrow \textcircled{1}$$

RHS: $\alpha T(x) + T(y)$

$$= \alpha \begin{bmatrix} 2a_{11} - a_{12} & a_{13} + 2a_{12} \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 2b_{11} - b_{12} & b_{13} - 2b_{12} \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2\alpha a_{11} - \alpha a_{12} & \alpha a_{13} + 2\alpha a_{12} \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 2b_{11} - b_{12} & b_{13} - 2b_{12} \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2(\alpha a_{11} + b_{11}) - \alpha a_{12} - b_{12} & \alpha a_{13} + b_{13} + 2(\alpha a_{12} + b_{12}) \\ 0 & 0 \end{bmatrix}$$

$$\text{LHS} = \text{RHS} \quad \therefore \begin{bmatrix} 2a_{11} - a_{12} & a_{13} + 2a_{12} \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2b_{11} - b_{12} & b_{13} - 2b_{12} \\ 0 & 0 \end{bmatrix}$$

To find: Basis of $N(T)$

$$N(T) = \{ A \in M_{2 \times 2}(\mathbb{R}) \mid T(A) = 0 \}$$

$$= \left\{ \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \mid \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \right\} \in M_{2 \times 2}(\mathbb{R})$$

$$= \left\{ \begin{bmatrix} 2a_{11} - a_{12} & a_{13} + 2a_{12} \\ 0 & 0 \end{bmatrix} \mid \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \right\}$$

$$2a_{11} - a_{12} = 0$$

$$a_{13} + 2a_{12} = 0$$

$$2a_{11} = \begin{bmatrix} a_{12} \\ 0 \end{bmatrix}$$

$$a_{13} = \begin{bmatrix} -2a_{12} \\ 0 \end{bmatrix} = -2 \begin{bmatrix} a_{12} \\ 0 \end{bmatrix} = -4a_{11}$$

$$= \left\{ \begin{bmatrix} a_{11} & 2a_{11} & -4a_{11} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \in M_{2 \times 3}(\mathbb{R}) \right\} = \left[(T) \right] \text{ with}$$

$$\text{Let } \begin{bmatrix} a_{11} & 2a_{11} & -4a_{11} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

$$= a_{11} \begin{bmatrix} 1 & 2 & -4 \\ 0 & 0 & 0 \end{bmatrix} + a_{21} \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} + a_{22} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} + a_{23} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Span } [N(T)] = \left\{ \begin{bmatrix} 1 & 2 & -4 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$$

$$\Rightarrow \dim [N(T)] = 4$$

$$\text{Nullity}(T) = 4$$

To find : Basis of $R(T)$

$$R(T) = \left\{ T(A) \in M_{2 \times 2}(\mathbb{R}) \mid A \in M_{2 \times 2}(\mathbb{R}) \right\}$$

$$= \left\{ \begin{bmatrix} 2a_{11} - a_{12} & a_{13} + 2a_{12} \\ 0 & 0 \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) \right\}$$

$$\text{Let } \begin{bmatrix} 2a_{11} - a_{12} & a_{13} + 2a_{12} \\ 0 & 0 \end{bmatrix}$$

$$= 2a_{11} - a_{12} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + (a_{13} + 2a_{12}) \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\text{span } [R(T)] = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right\}$$

$$\dim R(T) = 2 \Rightarrow \text{rank}(T) = 2$$

To verify: RND

$$\text{Rank}(T) + \text{Nullity}(T) = \dim [M_{2 \times 3}(\mathbb{R})]$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \begin{matrix} 2 + 4 = 6 \\ 6 = 6 \end{matrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ Hence it is verified.

To verify 1-1 & onto

$$\text{Here } N(T) \neq \{0\}$$

$$\therefore \dim [R(T)] = 2 \neq \dim [M_{2 \times 2}(\mathbb{R})] = 4$$

$\therefore T$ is not onto.

(iv) Prove that T defined as below is a linear transformation. Also find basis for $N(T)$ and $R(T)$. Compute nullity and rank of T and verify RND theorem. Also check if T is one-to-one, onto.

Define $T: P_2(\mathbb{R}) \rightarrow P_3(\mathbb{R})$ by $T(f(x)) =$

$$x f(x) + f'(x)$$

301: Given: $T = P_2(\mathbb{R}) \rightarrow P_3(\mathbb{R})$ by $T(f(x)) = xf(x) + f'(x)$
 $\forall f(x) \in P_2(\mathbb{R})$

To prove: T is linear

$$T[\alpha(f(x) + g(x))] = \alpha T[f(x)] + T[g(x)]$$

$$\therefore \text{L.H.S. } T[f(x)] = xf(x) + f'(x), \quad T[g(x)] = xg(x) + g'(x)$$

L.H.S:

$$\begin{aligned} T[\alpha(f(x) + g(x))] &= \alpha[xf(x) + f'(x) + xg(x) + g'(x)] \\ &= \alpha[x(f(x) + g(x)) + (f'(x) + g'(x))] \\ &= \alpha x(f(x) + g(x)) + \alpha(f'(x) + g'(x)) \\ &= \alpha x f(x) + \alpha x g(x) + \alpha f'(x) + \alpha g'(x) \quad \rightarrow \textcircled{1} \end{aligned}$$

R.H.S:

$$\begin{aligned} \alpha T[f(x)] + T[g(x)] &= \alpha[xf(x) + f'(x)] + xg(x) + g'(x) \\ &= [\alpha x f(x) + \alpha f'(x)] + xg(x) + g'(x) \\ &= \alpha x f(x) + \alpha f'(x) + xg(x) + g'(x) \end{aligned}$$

From $\textcircled{1}$ & $\textcircled{2}$ L.H.S = R.H.S

$\therefore T$ is linear transformation

To find: Basis for $P_2(\mathbb{R})$ & $P_3(\mathbb{R})$

$$P_2(\mathbb{R}) = \text{Span}\{1, x, x^2\}, \text{ so } \dim(P_2(\mathbb{R})) = 3.$$

$$P_3(\mathbb{R}) = \text{Span}\{1, x, x^2, x^3\}, \text{ so } \dim(P_3(\mathbb{R})) = 4$$

(i) To find on the basis elements of $P_2(\mathbb{R})$, compute T on the basis $\{1, x, x^2\}$

$$T(1) = x(1) + 0 = x + 0 = x$$

$$T(x) = x(x) + 1 = x^2 + 1$$

$$T(x^2) = x(x^2) + 2x = x^3 + 2x$$

To find Basis of $R(T)$

The range of T is spanned by

$$\{T(1), T(x), T(x^2)\} = \{x, x^2 + 1, x^3 + 2x\}$$

(i) Test for Linear Independence =

$$\text{Suppose } a(x) + b(x^2 + 1) + c(x^3 + 2x) = 0$$

$$\Rightarrow ax + bx^2 + b + cx^3 + 2cx = 0$$

Equating the like terms coefficient, we get

$$c = 0, b = 0, a + 2c = 0$$

$$\Rightarrow a = 0$$

(ii) It is linearly independent.

$$\text{The Range of } T = \{x, x^2 + 1, x^3 + 2x\}$$

$$\text{Rank}(T) = \dim R(T) = 3$$

To find Basis of $N(T)$

$$N(T) = \{f(x) \in P_2(\mathbb{R}) \mid T(f(x)) = 0\}$$

$$\text{Let } f(x) = a + bx + cx^2$$

$$\begin{aligned}\text{Then } T(f(x)) &= x(a + bx + cx^2) + b + 2cx \\ &= ax + bx^2 + cx^3 + b + 2cx \\ &= b + ax + 2cx + bx^2 + cx^3\end{aligned}$$

For $T[f(x)] = 0$, all coefficient must be zero

$$b = 0, c = 0, a = 0 \quad \therefore N(T) = \{0\}$$

$$\therefore \text{Nullity}(T) = \dim N(T) = 0$$

To verify RND

$$\text{Rank}(T) + \text{Nullity}(T) = \dim(P_2(\mathbb{R}))$$

$$3 + 0 = 3$$

$$3 = 3$$

$$\dim N(T) = 0$$

$$\therefore T \text{ is } 1-1$$

$$\dim(R(T)) = 3 \quad \dim[P_3(\mathbb{R})] = 4$$

$$\dim R(T) \neq \dim P_3(\mathbb{R})$$

$\therefore T$ is not onto.